

LECTURE 2 — ENERGY GENERATION

PHY199-RAP · Energy in the 21st Century

Dr. Salvatore Rappoccio



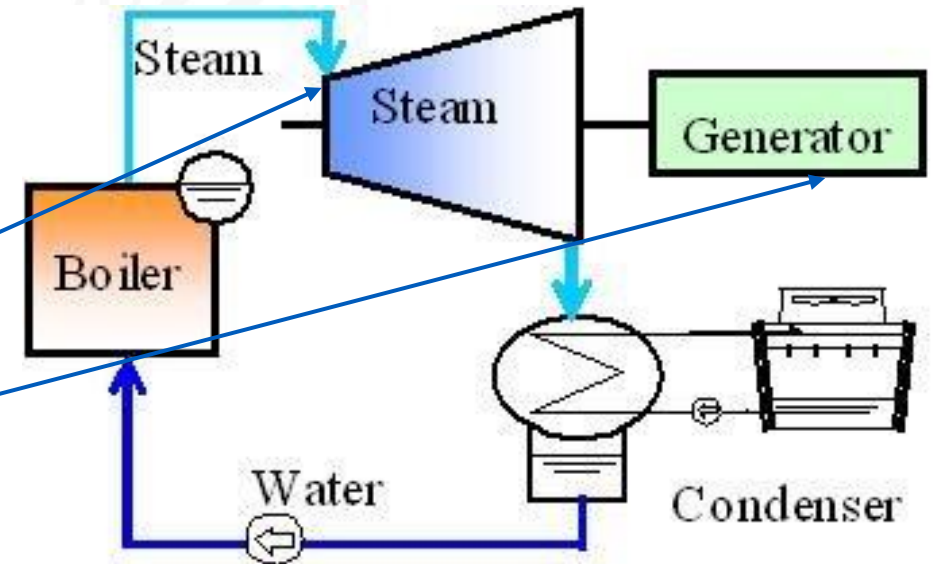
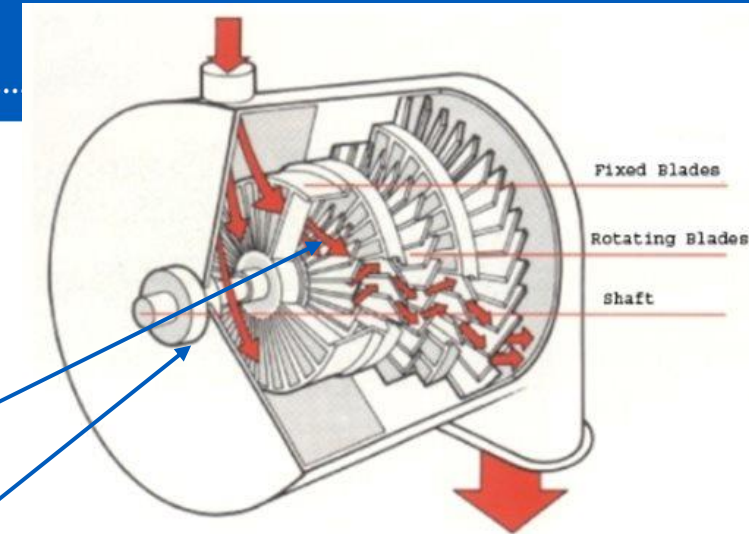
ELECTRICITY GENERATION : THE PHYSICS



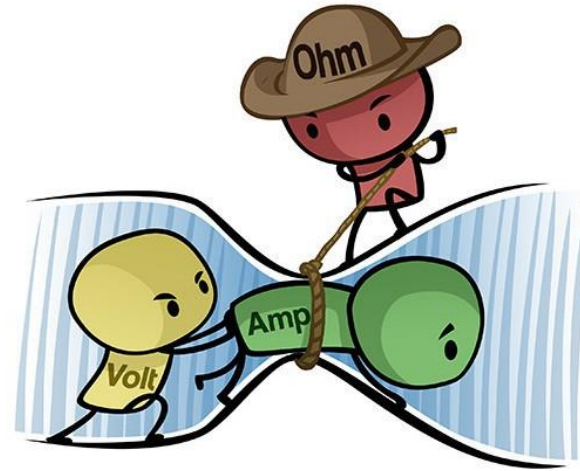
Turbines and Motors

- Turbine:

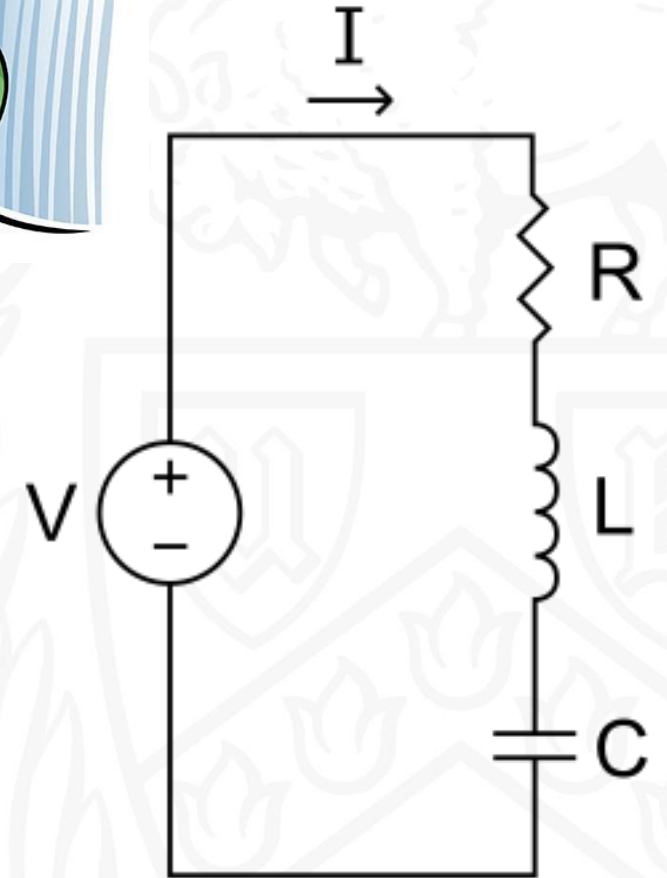
- Fixed blades and rotating blades are inside a casing.
- The rotating blades are attached to a shaft, which is then connected to a wire that rotates inside a magnetic field.
- Steam passes through, turning the rotating blades and the shaft, generating electricity



Circuits



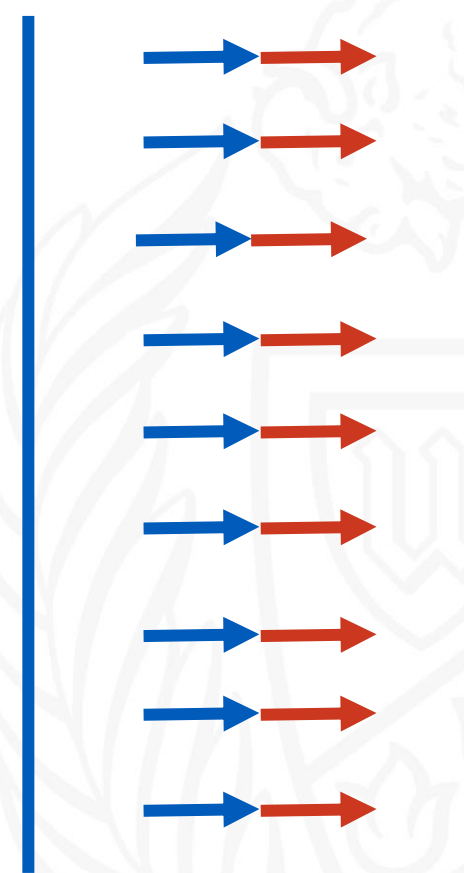
- Before we really understand power generation, we need to understand circuits
- We're going deeper into three concepts:
 - **Capacitance (C)**: stores energy in an electric field
 - **Inductance (L)** : stores energy in a magnetic field
 - **Resistance (R)** : dissipates thermal energy via heat



Now just draw a line to represent the charges, and you get a PARALLEL PLATE CAPACITOR

Electricity

- Teeny tiny bit of vector addition
- The components of Coulomb's law parallel to the plates CANCEL, the components perpendicular to the plates ADD, so you get a constant electric field
- Why is this important?
 - If you hook up a battery, this supplies a direct current electric field between the plates



Not responsible for this, just for enrichment

Capacitance

- If you put a positive and negative plate next to each other, it can **store electric energy**
- This is called a “**capacitor**” and how much it can store is the “**capacitance**”

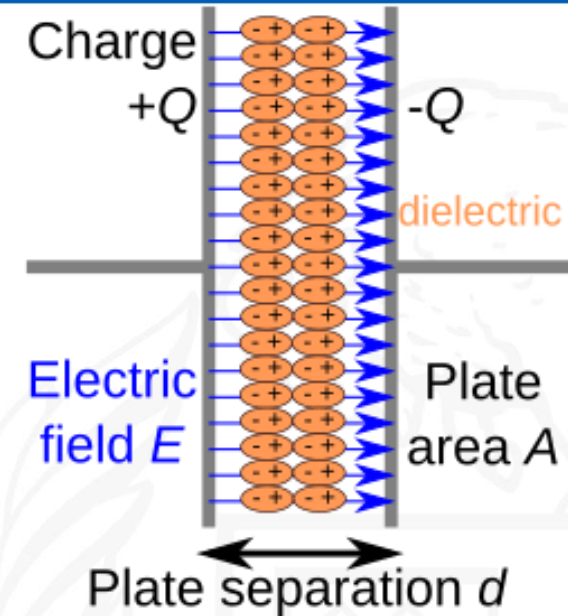
Capacitance:
1 Farad = 1
Coulomb / Volt

$$C = \frac{Q}{V}$$

Charge: Coulombs

You ARE responsible to understand
capacitance is charge divided by
voltage

Voltage: Volts



Induction

- Mathematically:
 - Electromotive force (voltage) is generated by changing magnetic field

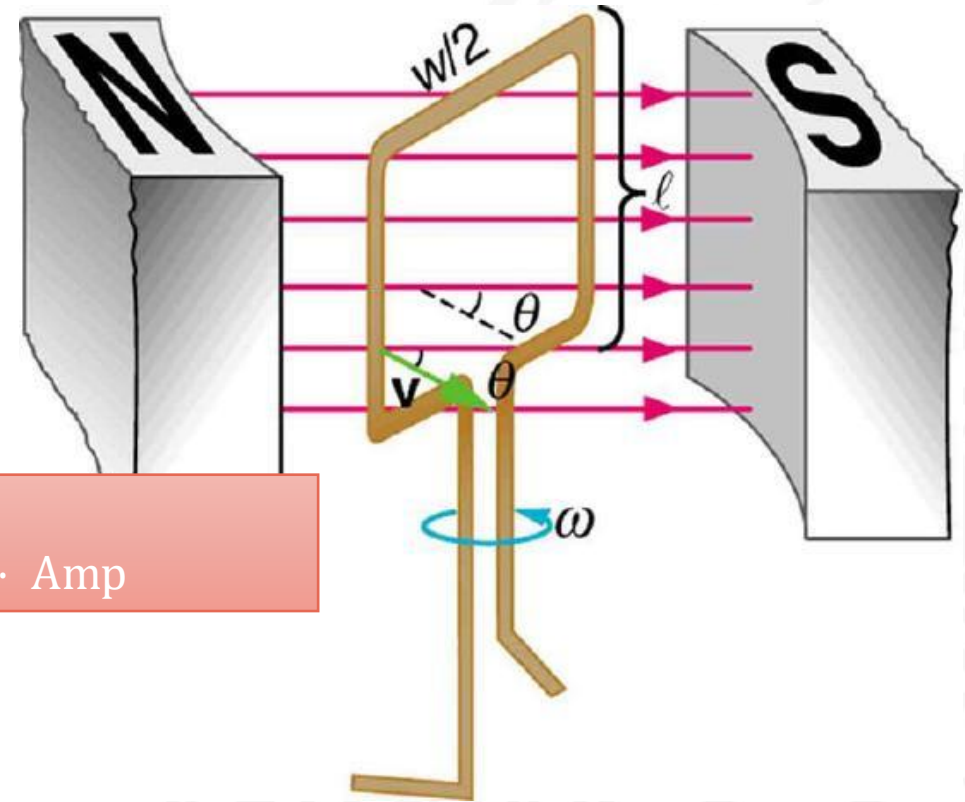
Electromotive force (Voltage):
1 Volt

Magnetic flux:
1 Weber = 1 Tesla · Amp

$$\mathcal{E} = \frac{\Delta\Phi_B}{\Delta t}$$

Time:
seconds

You ARE responsible to understand a voltage is induced from a changing magnetic flux



Inductance

- If you wrap a coil around something magnetic, it can **store magnetic energy** just like a capacitor stores electric energy
- This is called an “**inductor**” and how much magnetic flux it stores is called the “**inductance**”

Inductance:

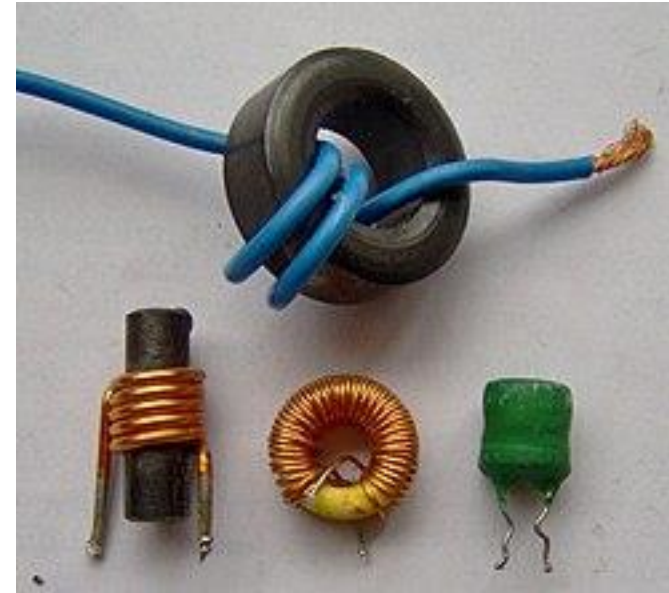
1 Henry = 1 Tesla x m² / Amp

Flux: Tesla x Area

$$L = \frac{\Phi_B}{i}$$

You ARE responsible to understand inductance is flux divided by current

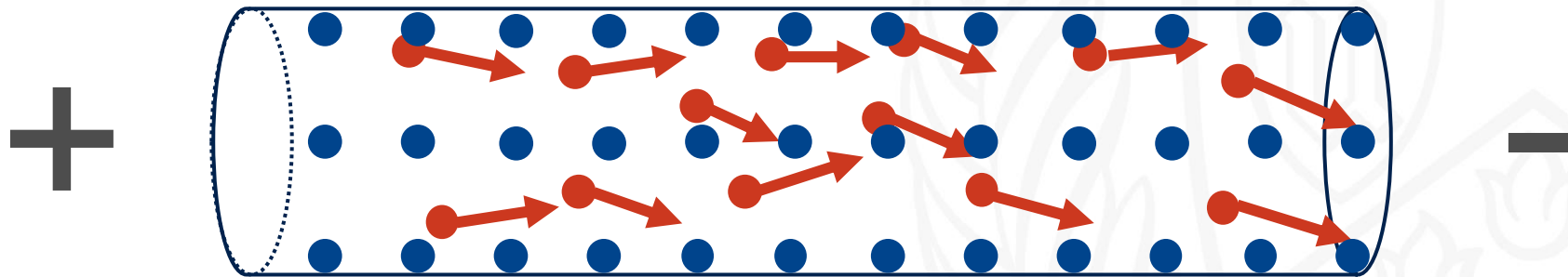
Current: Amps



Resistance

- Unfortunately, friction exists
- The electrons flowing through the wires bump into other electrons in the metal, so you get RESISTANCE
- Just like all resistance, it generates heat.

You ARE responsible to understand that resistance is voltage divided by current



Resistance:
Ohms = volts per amp

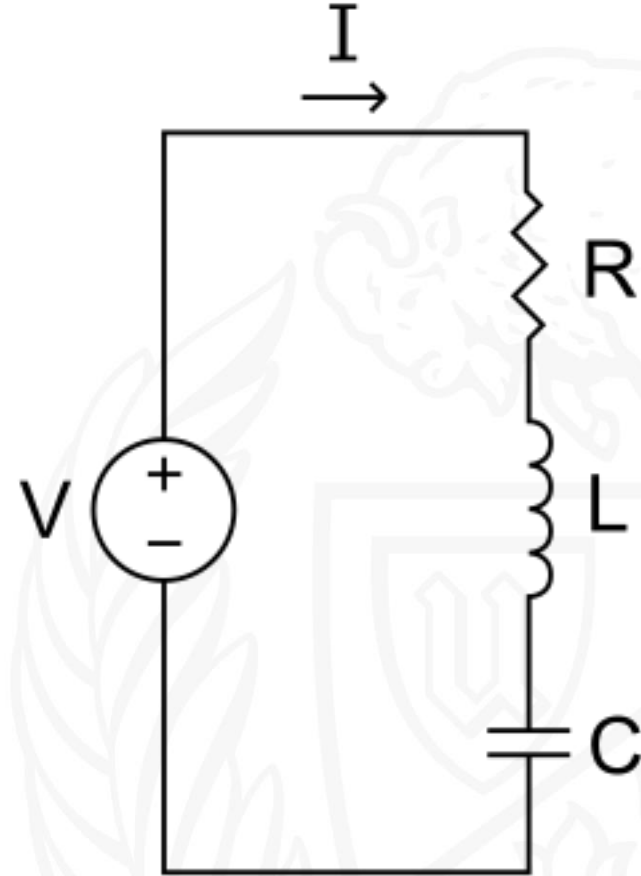
$$R = \frac{V}{i}$$

Electric potential:
Joules / Coulomb or **VOLTS**

Current:
Ampere = Coulombs / second

RLC Circuits

- What happens if you put a resistor, an inductor, and a capacitor in a circuit?
 - Capacitance stores electric energy
 - Inductor stores magnetic energy
 - Resistor heats up (but is what we're using for light bulbs, etc)



RLC Circuits

- This is a tough math problem so I'll just quote the answer, but for the math nerds (like me) the derivation is on the side (it's a second-order differential equation with damping!)
- Current will follow a **DAMPED COSINE WAVE**:

$$i(t) = i_0 e^{-\alpha t} \cos(2\pi f t)$$

Current:
Amps

Initial
current:
Amps

Base of Natural log
 $e = 2.71\dots$

Damping factor
 $\alpha = R/2L$

Cosine

Frequency
 $f \approx \frac{1}{2\pi\sqrt{LC}}$

$$V(t) = V_R(t) + V_C(t) + V_L(t)$$

$$V_R(t) = R i(t)$$

$$V_L(t) = L \frac{di}{dt}$$

$$V_C(t) = C \int i(t) dt$$

Suppose V is constant, take derivatives and divide by L :

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{RC} i(t) = 0$$

*Second order differential equation!
Solved with sine waves.*

Not responsible for this, just for enrichment

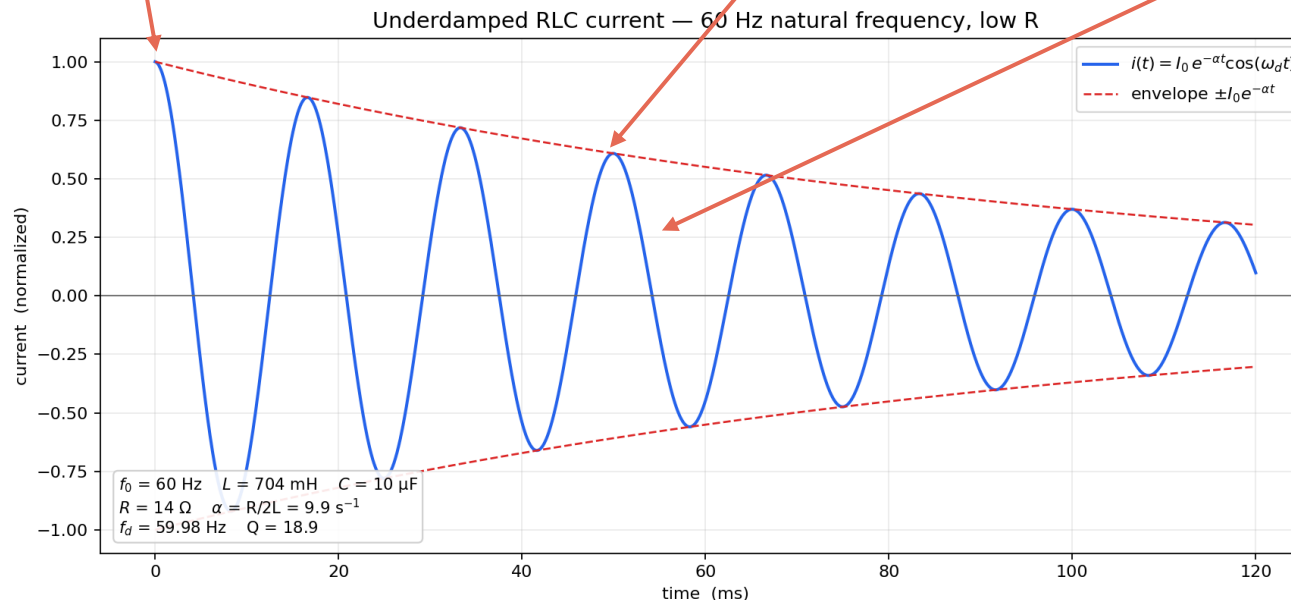
Current : Damped Cosine Wave

$$i(t) = i_0 e^{-\alpha t} \cos(2\pi f t)$$

Initial current

Exponential decay
from dissipative losses

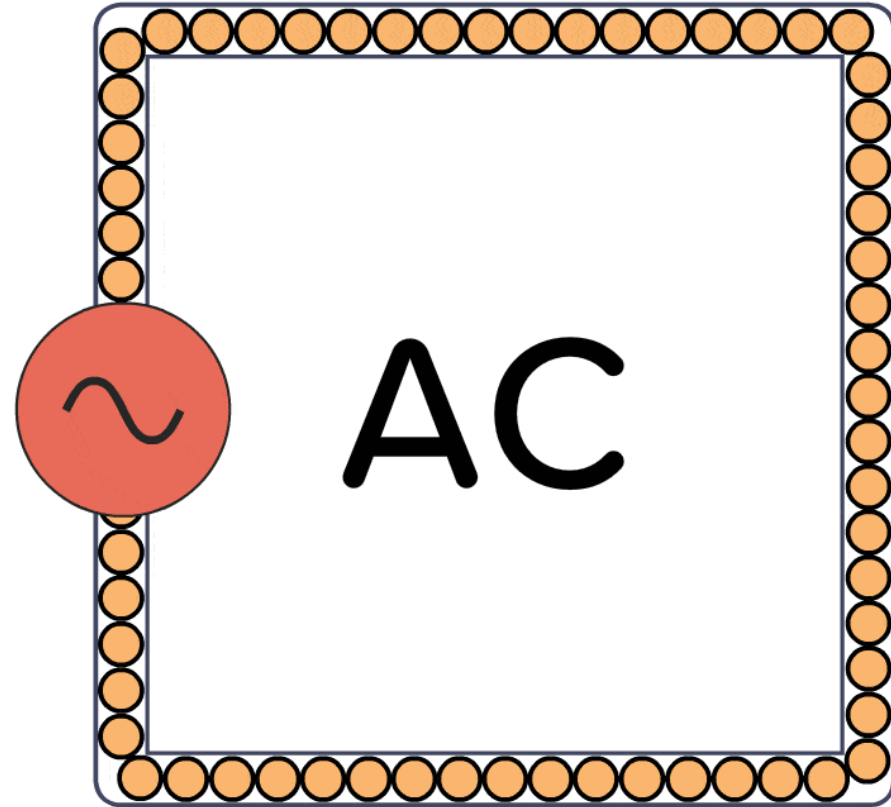
Cosine makes it wiggle at
frequency f



You ARE responsible to understand that an RLC circuit wiggles at a specific frequency and decays over time due to dissipative losses.

AC Current

- This is an **alternating current**
- The electrons in the material go back and forth, and can be used in powering circuits

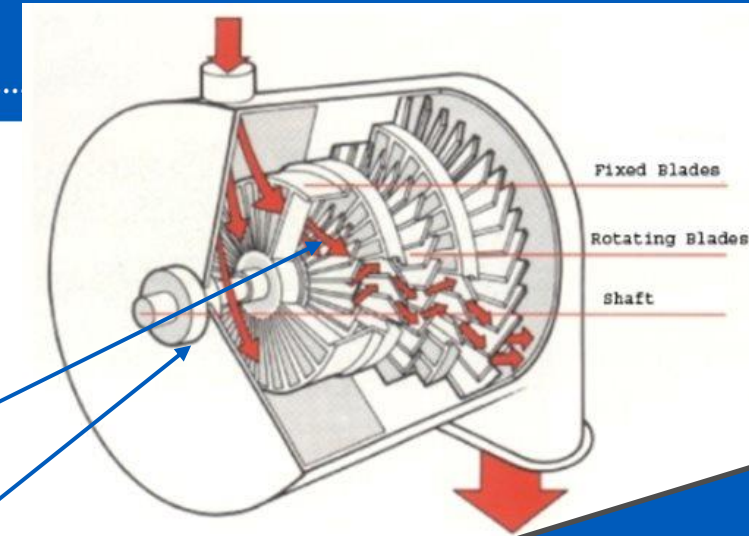


<https://www.build-electronic-circuits.com/difference-ac-vs-dc/>

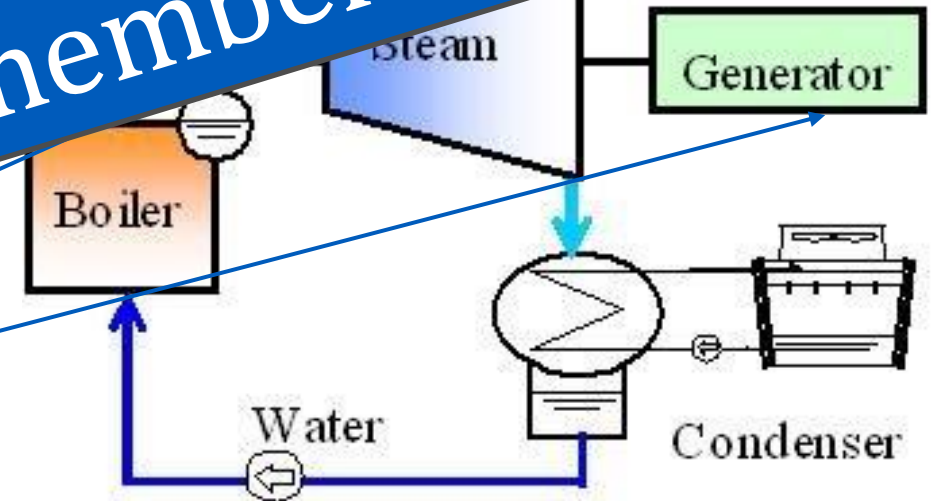
Turbines and Motors

- Turbine:

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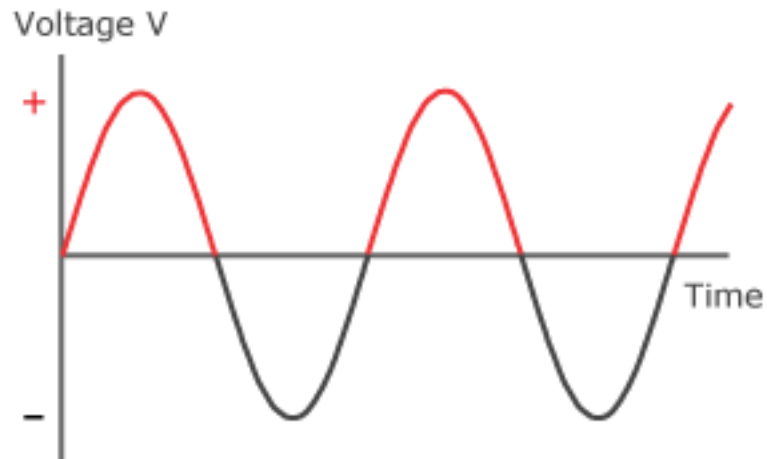


Remember turbines?

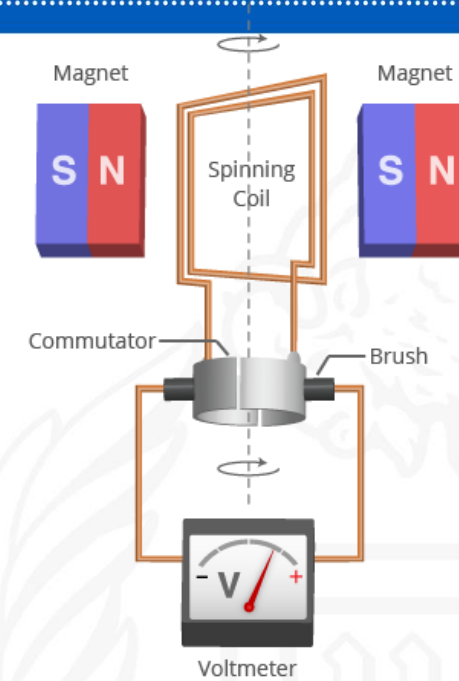
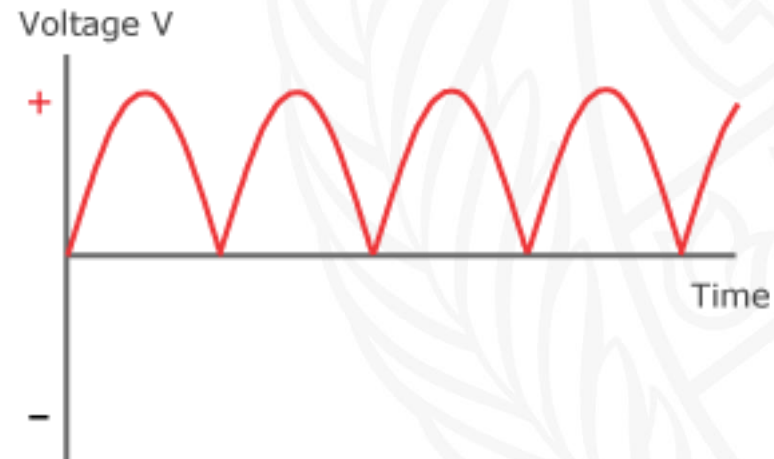


AC vs DC

- Turbines naturally produce AC power
- To convert to DC, it's necessary to make a commutator and use a dynamo that flips the sign every turn.
- More on this later!



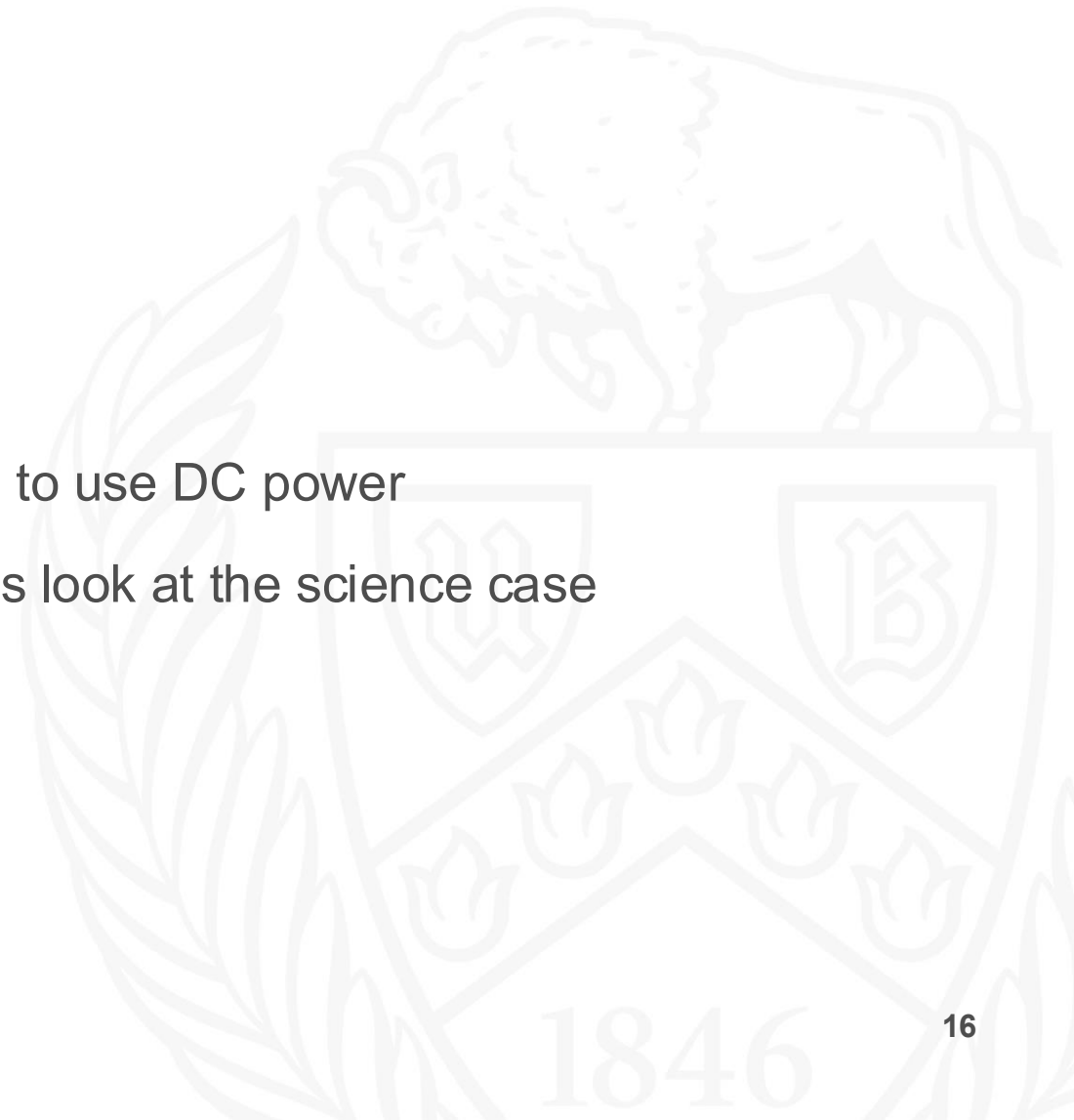
Flip - to +



<http://www.bigshotcamera.com/learn/power/dynamo>

AC vs DC

- Batteries were the first device used to generate power
- Smaller scale, can apply directly to what you need
- Batteries supply DC, so the first thoughts (Edison) were to use DC power
- We'll discuss the AC vs DC wars next class, but first let's look at the science case



Why AC won : Resistive Losses

- Remember resistance!

$$R = \frac{V}{i}$$

Resistance is voltage over current

- Or rewrite the equation:

$$V = iR$$

Voltage is current times resistance

- Power generated at the station is

$$P(\text{generated}) = iV$$

Power generated is current times voltage

- But remember the power **lost** by a resistor is

$$P(\text{lost}) = i^2 R$$

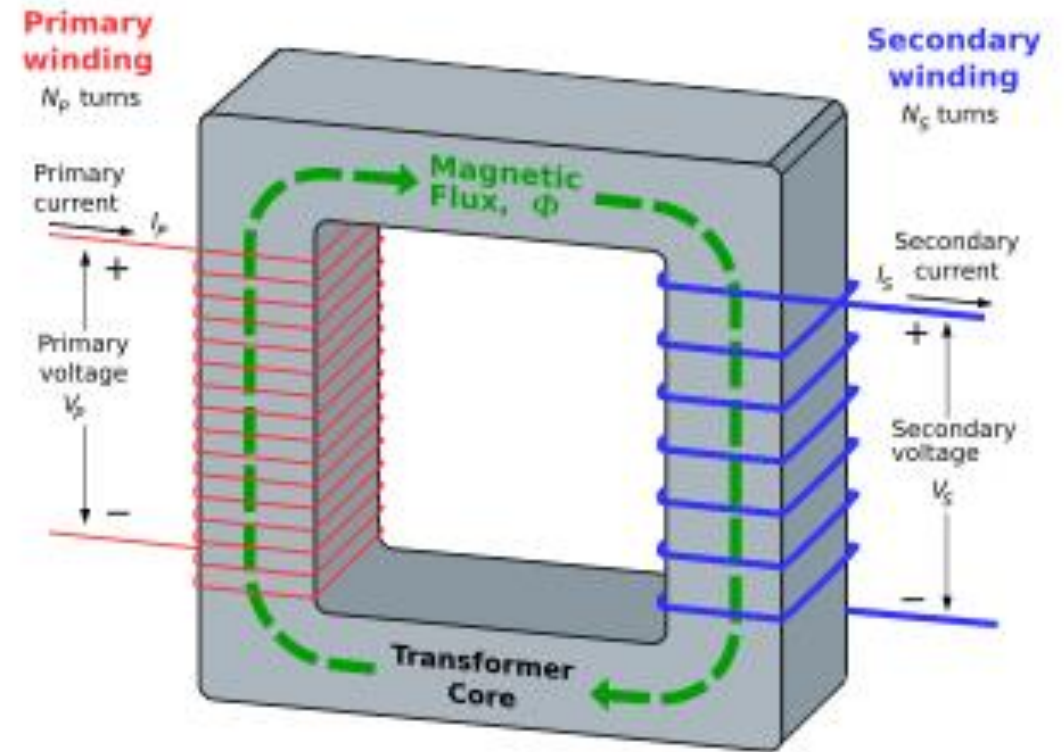
Power lost scales like current
SQUARED

Why AC Won: Transformers

- Now for the big technological advantage of AC current
- From Faraday's law:
- The ratio of voltages is:

$$V_1 = N_1 \frac{\Delta \Phi}{\Delta t}$$
$$V_2 = N_2 \frac{\Delta \Phi}{\Delta t}$$

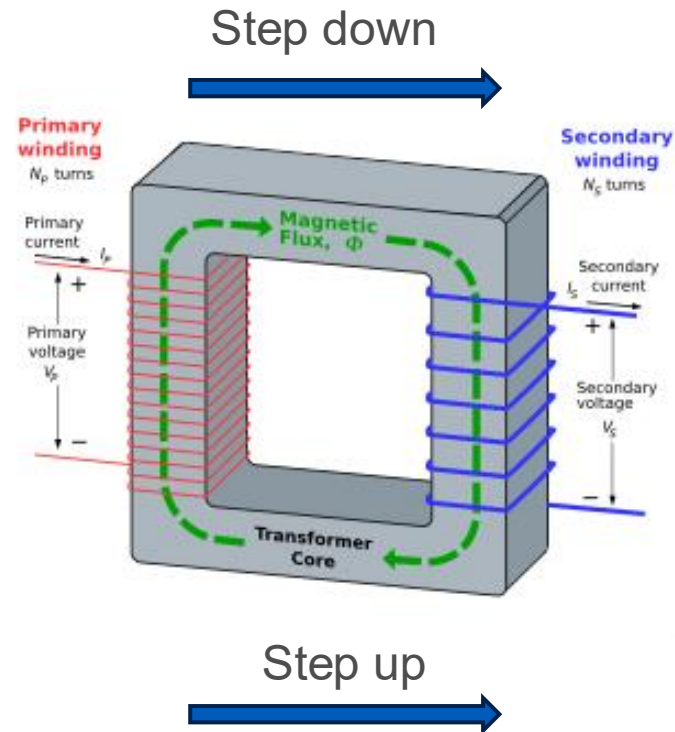
$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$



<https://en.wikipedia.org/wiki/Transformer>

Why AC Won: Transformers

- Transformers can **STEP UP** or **STEP DOWN** voltage



You ARE responsible to understand that you can use a transformer to step voltage up or down

Tophat question

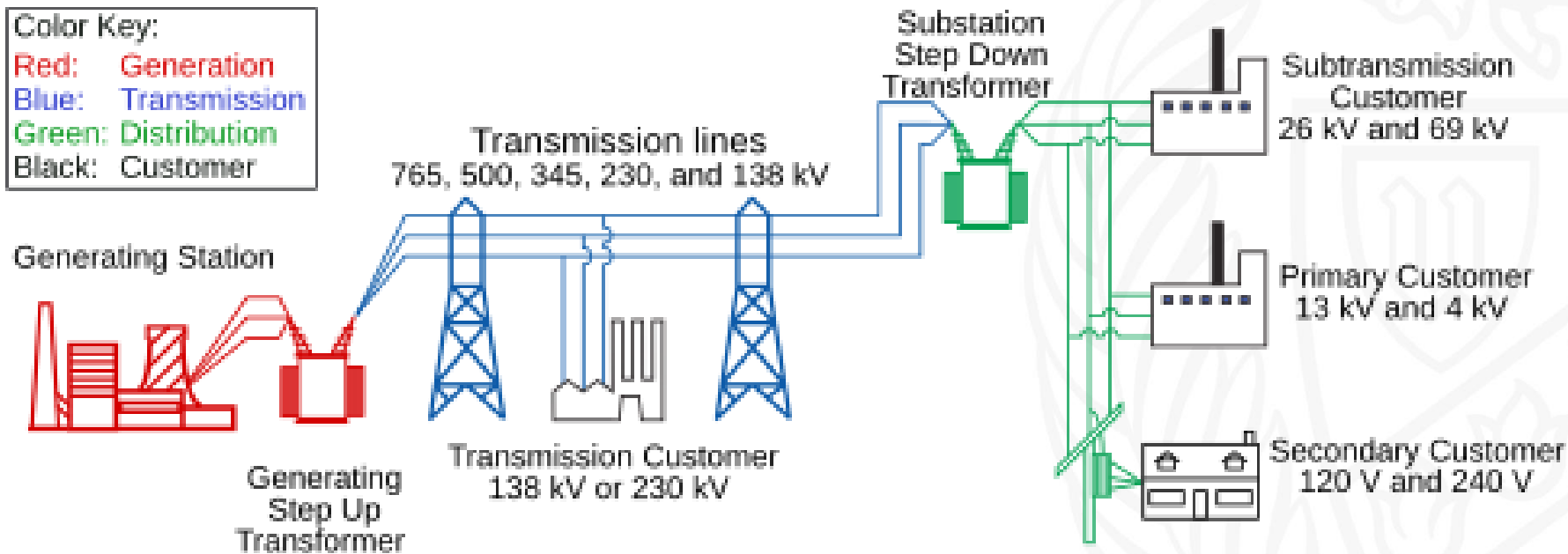
- If you have a constant power source with $P(\text{generated}) = iV$, and the power lost has $P(\text{lost}) = i^2 R$, is more power lost at high voltage, low voltage, or they are the same?
 - A: More power lost at high voltage.
 - B: More power lost at low voltage.
 - C: The power losses are the same.

Tophat question

- If you have a constant power source with $P(\text{generated}) = iV$, and the power lost has $P(\text{lost}) = i^2 R$, is more power lost at high voltage, low voltage, or they are the same?
 - A: More power lost at high voltage.
 - B: More power lost at low voltage.
 - C: The power losses are the same.

You ARE responsible to understand / remember that for a constant power generated, there are more losses at low voltage than at high voltage

Now You Understand The Grid



To remember

Concept (need to remember)	Formula (for enrichment)
capacitance is charge divided by voltage	$C = \frac{Q}{V}$
a voltage is induced from a changing magnetic flux	$\mathcal{E} = \frac{\Delta\Phi_B}{\Delta t}$
inductance is flux divided by current	$L = \frac{\Phi_B}{i}$
resistance is voltage divided by current	$R = \frac{V}{i}$
Alternating current (AC) circuits wiggles at a specific frequency and decays over time due to dissipative losses.	$i(t) = i_0 e^{-\alpha t} \cos(2\pi f t)$
you can use a transformer to step voltage up or down	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$
for a constant power generated, there are more losses at low voltage than at high voltage	$P(\text{lost}) = i^2 R$

Links & References

References

[Steam turbines](#)

[Capacitor](#)

[Inductor](#)

[RLC circuit](#)

[AC vs DC](#)

[Dynamo \(BigShot\)](#)

[Commutator](#)

[Dynamo \(Wikipedia\)](#)

[Transformer](#)

[Power transmission](#)

Image credits (Wikimedia Commons)

[Capacitor image 1](#)

[Capacitor image 2](#)

[Inductor image 1](#)

[Inductor image 2](#)

[Transformer image](#)

